

Sizing Rubble-Mound Armour Units

The Hudson (SPM 1984) stability formula and the Van der Meer (1988) rock formulae, with worked example. Companion note to the online Armour Stone Sizing calculator.

1. The stability problem

The primary armour of a rubble-mound breakwater or revetment must be heavy enough to resist displacement by wave action. Two methods dominate practice: the empirical **Hudson** formula, simple and applicable to rock and concrete units through a stability coefficient K_D ; and the **Van der Meer** formulae for rock, which additionally account for wave period, storm duration, permeability and an accepted damage level. Both are framed around the relative buoyant density $\Delta = \rho_r/\rho_w - 1$ and the nominal diameter $D_{n50} = (M_{50}/\rho_r)^{1/3}$.

2. Hudson formula (SPM 1984)

$$M_{50} = \rho_r \cdot H^3 / (K_D \cdot \Delta^3 \cdot \cot \alpha), \quad W = M_{50} \cdot g, \quad D_{n50} = (M_{50}/\rho_r)^{1/3}.$$

Here H is the design wave height (the SPM recommends $H = H_{1/10} \approx 1.27 \cdot H_s$ for the no-damage condition), α the seaward slope, and K_D the stability coefficient, which depends on the armour unit and whether waves break on the structure:

Armour unit (trunk, random placement)	K_D breaking	K_D non-breaking
Rough angular quarystone, 2 layers	2.0	4.0
Smooth rounded quarystone, 2 layers	1.2	2.4
Modified cube (concrete), 2 layers	6.5	7.5
Tetrapod (concrete), 2 layers	7.0	8.0
Dolos (concrete), 2 layers	15.8	31.8
Accropode (concrete), 1 layer	~12	~15

3. Van der Meer formulae (1988, rock)

For rock armour in deep water, the required stability number $N_s = H_s/(\Delta D_{n50})$ depends on the breaker regime, set by the Iribarren number $\xi_m = \tan \alpha / \sqrt{s_m}$ (with the mean period, $s_m = 2\pi H_s / g T_m^2$):

$$\text{Plunging } (\xi_m < \xi_{cr}): H_s/(\Delta D_{n50}) = 6.2 \cdot P^{0.18} \cdot (S/\sqrt{N})^{0.2} \cdot \xi_m^{-0.5}.$$

$$\text{Surging } (\xi_m \geq \xi_{cr}): H_s/(\Delta D_{n50}) = 1.0 \cdot P^{-0.13} \cdot (S/\sqrt{N})^{0.2} \cdot \sqrt{(\cot \alpha) \cdot \xi_m^P}, \quad \xi_{cr} = [6.2 \cdot P^{0.31} \sqrt{\tan \alpha}]^{1/(P+0.5)}.$$

P is the notional permeability (0.1 impermeable core, 0.4 permeable core, 0.5 homogeneous, 0.6 no core); S the damage level ($S = 2$ start of damage, 3–5 intermediate, 8–17 failure depending on slope); and N the number of waves in the design storm (typically 1000–7000). D_{n50} then follows from $D_{n50} = H_s/(\Delta \cdot N_s)$ and $M_{50} = \rho_r D_{n50}^3$.

4. Design curves

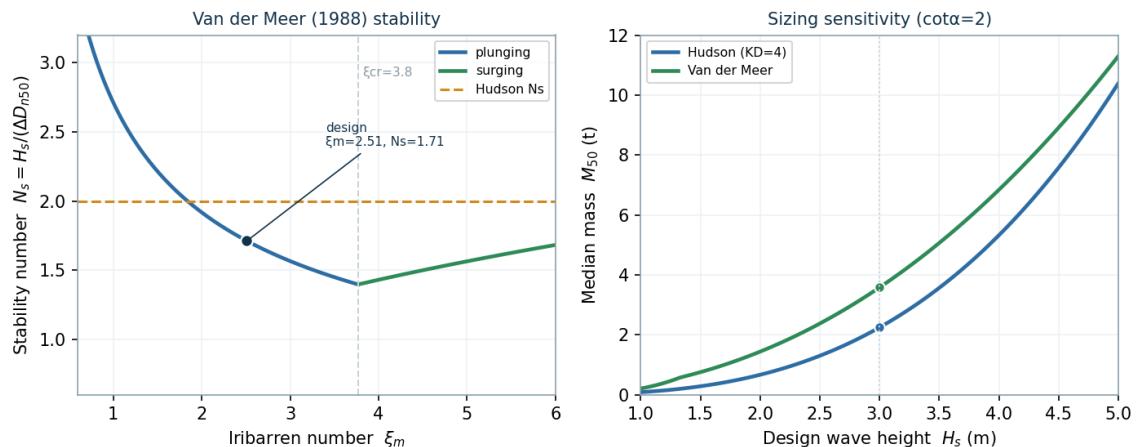


Figure 1. Left: Van der Meer stability number versus Iribarren number — the plunging and surging branches meet at ξ_{cr} , where N_s is minimum (largest stone). Right: median mass versus design wave height for the worked example ($\cot \alpha = 2$); Van der Meer governs over Hudson here.

5. From D_{n50} to the layer

The armour layer thickness for n layers is $t_a = n \cdot k_\Delta \cdot D_{n50}$ (layer coefficient $k_\Delta \approx 1.0\text{--}1.15$ for rock), and the number of units per unit area is $N_a/A = n \cdot k_\Delta \cdot (1 - n_v) \cdot D_{n50}^{-2}$, with layer porosity $n_v \approx 0.37\text{--}0.40$ for rock. Where both methods apply, the larger stone governs; Van der Meer is usually more conservative because it targets a defined damage level over a full storm.

6. Worked example

Quantity	Value
Inputs	$H_s = 3.0$ m, $T_p = 8.0$ s, $\cot\alpha = 2$, rough angular rock (non-breaking), $P = 0.4$, $S = 2$, $N = 3000$
Relative density, Δ	1.585 ($\rho_r = 2650$, $\rho_w = 1025$ kg/m ³)
Hudson ($K_D = 4.0$)	$M_{50} = 2.24$ t, $D_{n50} = 0.95$ m, $W = 22.0$ kN
Armour layer / placement	$t_a = 2.18$ m, ~ 162 units per 100 m ²
Van der Meer regime	$\xi_m = 2.51 < \xi_{cr} = 3.77 \rightarrow$ plunging
Van der Meer, $N_s = 1.71$	$M_{50} = 3.58$ t, $D_{n50} = 1.11$ m
Governing design	$M_{50} = 3.58$ t (Van der Meer, larger)

Notation

H_s significant wave height; H design wave height; M_{50} median armour mass; W armour weight; D_{n50} nominal diameter; ρ_r , ρ_w armour and water density; Δ relative buoyant density; K_D Hudson stability coefficient; $\cot\alpha$ slope; N_s stability number; ξ_m Iribarren number; P notional permeability; S damage level; N number of waves; n number of layers; k_Δ layer coefficient.

References

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